

An efficient and physics-informed regional maize yield estimation scheme by combining data assimilation and machine learning

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ABSTRACT

Accurate regional crop yield prediction is essential for agricultural development and food security. Data assimilation (DA) frameworks combined with satellite observations improve regional yield estimation but remain computationally intensive at large scales. This study aims to develop an efficient, physics-informed regional yield estimation framework by integrating DA with machine learning (ML) to reduce computational costs and improve accuracy. An iterative ensemble smoother DA algorithm was applied with SWAP crop growth model to generate “virtual” yield observations across randomly selected maize pixels. These observations, together with remote sensing data, were used to train four ML models—Feature Tokenizer Transformer (FTT), Artificial Neural Network (ANN), eXtreme Gradient Boosting (XGBoost), and Random Forest (RF)—for regional maize yield estimation in Shandong Province, China. The results reveal that the DA-ML framework significantly reduces computational costs compared to traditional pixel-based DA methods (i.e., achieving a computational time saving of over 99.8 %), while maintaining comparable yield estimation accuracy. Among the model, RF achieved the best performance, with regional yield estimates across 114 counties showing $R^2 = 0.62$, $RMSE = 1.19 \times 10^5$ t in the calibration year and $R^2 = 0.49$, $RMSE = 1.28 \times 10^5$ t in the validation year. Furthermore, relative importance analysis from the DA-ML framework indicates that radiation, soil moisture, and rainfall during the jointing stage are the most influential variables for regional maize yield estimation, followed by wind speed, air humidity, and temperature during the reproductive stage, highlighting the critical role of environmental conditions in the later growth stages for determining final yield outcomes. This study showcases a reliable DA-ML system for rapid-responsive regional crop yield estimation and management decisions, with potential applicability to other crops and regions.

1. Introduction

Timely and precise regional crop monitoring and yield prediction are essential for agricultural development and food security. Crop growth models, using mathematical equations to characterize key physical processes of crops, provide a valuable tool to simulate crop growth and forecast crop yield (Rauff and Bello, 2015). However, implementing crop models over large areas is difficult, since the required inputs, including soil properties, crop types, and meteorology conditions, are significantly heterogeneous and difficult to obtain on the regional scales

(Hansen and Jones, 2000). In recent decades, advances in remote sensing (RS) technology have enabled large-scale soil and crop monitoring (Karthikeyan et al., 2020). Thus, incorporating RS measurements into crop growth models could help deal with spatial heterogeneity and enhance the accuracy of crop growth modeling.

Over the past fifty years, numerous satellite missions have been launched to provide multi-source observations for crop growth research. For example, optical imageries from Sentinel-2, Landsat, and MODIS (Moderate Resolution Imaging Spectroradiometer) have been employed to derive essential vegetation parameters (Dimitrov et al., 2019), such as

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leaf area index (LAI), fractional cover (fCOVER), fraction of photosynthetically active absorbed radiation (fAPAR), and plant chlorophyll content. Besides, microwave missions (e.g., Soil Moisture Active Passive, SMAP) facilitate surface soil moisture (SSM) retrieval. Moreover, a variety of reanalysis datasets (e.g., ERA5) offers critical hydro-climatic data (e.g., radiation, humidity, air temperature, and rainfall) based on global satellite and ground measurements (Hersbach et al., 2020), which can be used in coupled hydrological-agricultural applications.

Data assimilation (DA) is a useful tool to correctly describe an agroecosystem state by integrating information from physical models and remote sensing observations (Jin et al., 2018). Plenty of studies utilized DA methods to incorporate multi-source data into crop models for enhanced agroecosystem simulations (Botto et al., 2018). For instance, the satellite-based LAI and SSM were integrated into a crop growth model by Ines et al. (2013), which illustrated that a combination of LAI and SSM data enhanced the maize yield simulations compared to assimilating either LAI or SSM data individually. However, the utilities of DA methods are still limited in regional yield estimation due to two main challenges: (1) The inherent structure of physical models. Challenges exist in incorporating state-of-the-art RS observations (e.g., hyperspectral) (Sahoo et al., 2015) or solar-induced chlorophyll fluorescence (Sun et al., 2017) into the traditional crop growth model (Yu et al., 2020); (2) Computational efficiency. DA algorithms require considerable computational costs, making it difficult to be applied over large areas with thousands even millions of computational grids (Dorigo et al., 2007).

Compared to DA methods, machine learning (ML) can establish non-linear relationships between multi-source data and crop yield with a low computational cost (Chlingaryan et al., 2018), providing a promising way to forecast crop yield on large scales (Salvador et al., 2020; Han et al., 2021). However, the prediction accuracy of ML approaches is highly related to the data quantity and quality (Chlingaryan et al., 2018). Most ML methods are unable to extract interpretable information and knowledge from the vast streams of data (Kashinath et al., 2021). Moreover, purely data-driven ML models may fit observations very well, but due to extrapolation or observational biases, their predictions can be inconsistent with physical laws or lack credibility, resulting in poor generalization (Bagheri et al., 2024). Therefore, there is an urgent need to integrate fundamental physical laws and domain knowledge into ML models by “teaching” them relevant physical principles.

Physics-informed machine learning, as defined by Karniadakis et al. (2021), indicates the process of enhancing learning algorithms with prior knowledge derived from our observations, experience, and understanding of physics or mathematics. This ensures that the resulting models not only fit data but also respect the governing physics, thereby improving generalization and interpretability. A simple and practical physics-informed way is to employ outputs of the physical models as the substitute for actual observations to train ML models, which are also known as “emulators” (Blanc, 2017; Ramankutty et al., 2013). For example, the MCYFS yield data (produced by the WOFOST model) has been used as the target variable in the ML model to forecast the yield of six crop types in Europe (Paudel et al., 2022, Paudel et al., 2021). Similarly, the maize yield and nitrogen loss simulated by APSIM model were employed to train ML algorithms (Shahhosseini et al., 2019). These synthetic data (i.e., outputs of the physical models) could reflect the fundamental physical laws, and therefore are able to introduce a weak but valuable physical mechanism into an ML model in its training stage (Karniadakis et al., 2021). However, forward physical models (e.g., WOFOST model and APSIM model) are unable to directly integrate key crop growth variables (e.g., LAI and SSM) into the modeling process, deteriorating the accuracy of model outputs.

DA-enhanced outputs, on the other hand, reflect both physical mechanisms and observational data, making them more suitable for training physics-informed ML models. Recently, Zhuang et al. (2024) and Lu et al. (2025) employed DA outputs to train ML models for estimating regional yield of winter wheat and rice, respectively, and

demonstrated notable improvements in yield prediction accuracy. However, these studies rarely address spatial uncertainties associated with initial conditions and crop growth parameters, which may introduce biases into regional yield estimation. Furthermore, “mixed pixels” (i.e., spatial heterogeneity of crops within a grid cell) play significant effects on DA outputs, yet these effects remain insufficiently explored.

Therefore, this study aims to combine data assimilation and machine learning approaches to achieve an efficient and physics-informed regional yield (or parameter) estimation scheme. The specific goals are to: (1) analyze the global sensitivity of parameters in a crop growth model—Soil-Water-Atmosphere-Plant (SWAP) model—for LAI and SSM, and identify the most critical spatially-variable parameters for inverse modeling within the regional area; (2) establish four advanced machine learning models based on DA outputs and RS data, and assess the effectiveness of these emulators; (3) evaluate and validate the accuracy of regional yield estimation generated by four ML emulators.

2. Data

2.1. Study area

The study area is subjected to the whole Shandong Province of China (34°25' E to 38°23' E, 114°36' N to 122°43' N in Fig. 1). This region is part of the North China Plain and covers approximately 157,900 km². In this area, the most common cropping pattern is the rotation of summer maize with winter wheat. The average annual temperature is from 12 °C to 14 °C, and the average annual precipitation ranges from 600 to 900 mm. Notably, a dedicated long-term observation site has been established at the Yucheng Comprehensive Experiment Station (36.83°N, 116.57°E) to systematically gather field crop growth and meteorological data in this region (Fig. 1).

Summer maize is typically sown in the middle of June, and harvested at the end of September in Shandong. Thus, the satellite images were collected from 18th June to 30th September in 2019 and 2020, specifically from the day of year (DOY) 169 (170) to DOY 273 (274) in 2019 (2020). Among them, the site-level data from seven plots were collected for model calibration and validation, while the official statistics from 114 counties were utilized to validate the results at the province level.

2.2. Site data

At the Yucheng Station, measurements were collected in 2019 and 2020 from seven experimental plots with sufficient nitrogen and irrigation treatments for summer maize (cultivar “Zhengdan 958”) (Yu et al., 2023), as presented in Fig. S1 in the supplementary material. This two-year site data was used for model calibration and site-level assessment. LAI was collected utilizing an LAI-2200 Plant Canopy Analyzer in the experimental plots. Volumetric soil moisture was manually observed at depths of 10 cm, 20 cm, 40 cm, 60 cm and 80 cm using Frequency Domain Reflectometry (FDR) with the IMKO TRIME®-PICO-IPH device. In this study, the soil moisture data at a depth of 10 cm were selected to approximate the SSM. Additionally, the maize yield data are also collected. The maize kernels were heated at 105 °C for two hours, and dried at 75 °C until a constant weight was achieved.

In 2020, LAI and SSM were in-situ measured throughout the entire growth period on the following dates (mm/dd): 06/28, 07/07, 07/16, 07/24, 08/01, 08/09, 08/14, 08/26, 09/11, and 09/29. The maize grain yield was collected on 10/05. In 2019, the SSM data were not monitored. LAI data were observed on 07/19, 08/05, and 08/22, with yield data collected on 10/05. Given the greater abundance of data from 2020, we utilized the 2020 dataset for model calibration and the 2019 dataset for validation. In addition, an automated weather station set up at the site recorded long-term meteorological data. The agricultural management practices (i.e., irrigation and fertilization) were also recorded at Yucheng Station.

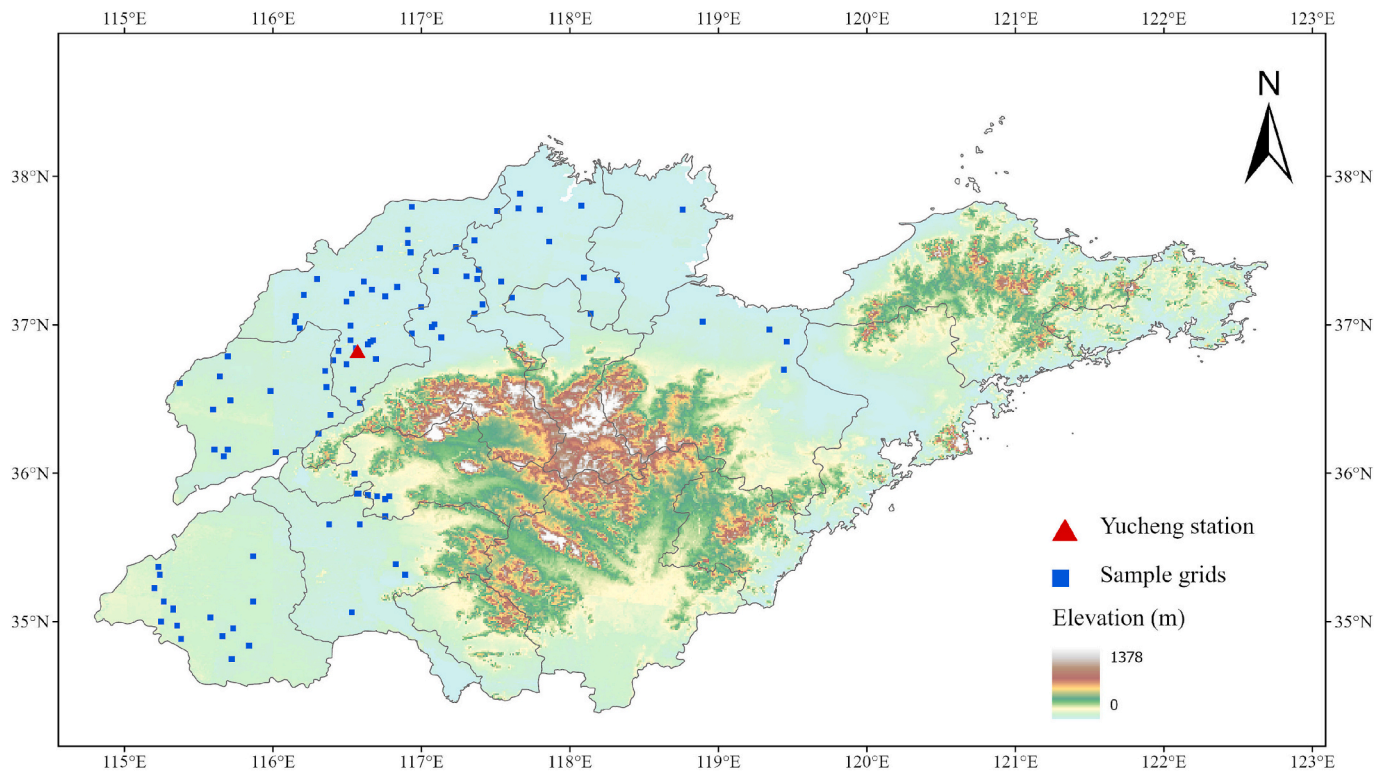


Fig. 1. Study area in Shandong Province, China. The red triangle indicates a long-term observation site (i.e., Yucheng Comprehensive Experiment Station). The blue squares represent 100 randomly-selected pixels simulated at a 1-km scale, as described in Section 3.3.2.

2.3. Regional-scale data

In this work, meteorological data and crop-growth related data were collected and processed on the Google Earth Engine (GEE) platform. First, Sentinel-2 Level-2A (L2A) reflectance dataset are utilized to generate a 10-m crop cultivation map for summer maize (details for mapping maize growing area are presented in Section 2.4). Second, MODIS LAI (MOD15A2H V6.1) and reflectance (MOD09A1 V6.1) data were obtained to characterize vegetation vigor and moisture content of maize canopy. Three widely-used VIs, namely the normalized difference vegetation index (NDVI), the normalized difference water index (NDWI), and the enhanced vegetation index (EVI) were calculated from MODIS reflectance data to characterize vegetation vigor and moisture content of maize canopy (Eqs. S1-S3). In addition, daily meteorological data from the ERA5-Land reanalysis products are used as the input variables for both crop growth and machine learning models. Finally, the SSM at the depth of 5 cm is obtained from the SMAP data. Due to the significant systematic bias between SMAP SSM and actual soil moisture data, the regional SSM data are calibrated using a linear regression equation which is established by the relationship between SMAP and site SSM measured at the Yucheng Station (Colliander et al., 2017). The average value of MODIS LAI and calibrated SMAP SSM over the whole maize growth stage across Shandong Province are presented in Fig. S2.

Table 1 presents a summary of these data. The detailed descriptions of each data source and extracted variables are presented in the Data S1 in the supplementary materials.

In addition, agricultural statistical data sourced from the China Statistical Yearbook (<https://tjj.shandong.gov.cn/col/col322452/index.html>) are utilized to assess the model accuracy for regional yield and planting area. The datasets comprise summer maize area and yield records from 114 counties in 13 municipal districts in Shandong Province for the years 2019 and 2020, which are provided in Table S1.

Table 1

A brief summary of regional-scale data for model inputs.

Types	Variables	Spatial resolution	Temporal Resolution	Source
Crop data	Crop cultivation map	10 m	—	Sentinel-2
	LAI (m^2/m^2)	500 m	8-day	MODIS
Vegetation indices	NDVI	500 m	8-day	MODIS
	EVI			
Meteorology	NDWI			
	Radiation (kJ/m^2)	11 km	Daily	ERA5
	Air temperature ($^{\circ}\text{C}$)			
	Dew point temperature ($^{\circ}\text{C}$)			
	Wind speed (m/s)			
Soil	Rainfall (mm)			
	Surface soil moisture (m^3/m^3)	10 km	3-day	SMAP

2.4. Mapping 10-m maize cultivation

Summer maize classification maps in 2019 and 2020 were generated using 10-m Sentinel-2 reflectance images, as the most recent high spatial-resolution maize cultivation map has not been produced yet. The classification method follows the approach outlined by Saini and Ghosh (2018), with a detailed description of the process provided in Data S1 (Section 3.5).

The 10-m maize cultivation maps for the years 2019 and 2020 are presented in Fig. 2(a) and (c). We further computed the total area of maize cultivation by summing the pixels within each administrative unit for comparative analysis with statistical surveys, as shown in Fig. 2(b) and (d). The results show that the coefficient of determination (R^2) between the estimated maize area and official statistic data in 2019 and 2020 are 0.57 and 0.73, with a slope value of 0.70 and 0.77,

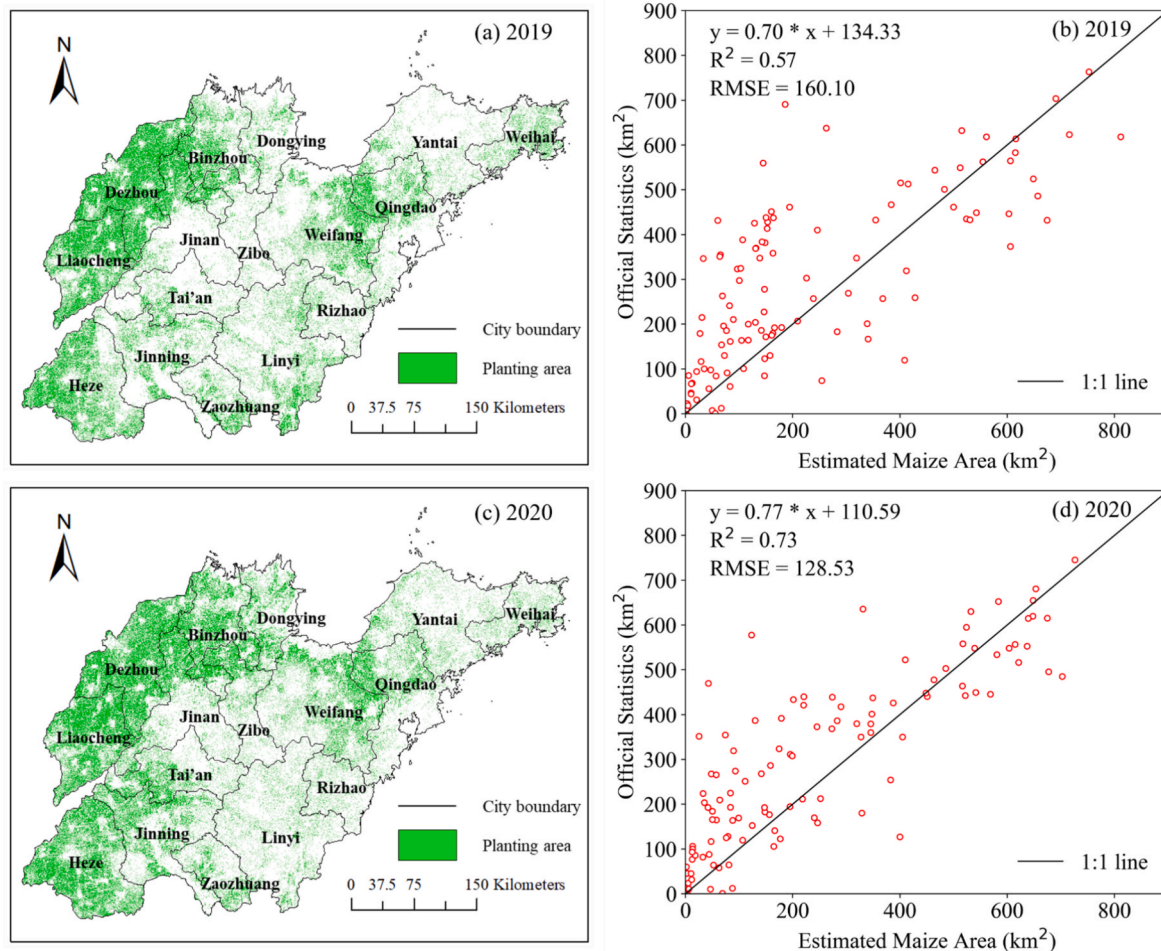


Fig. 2. Map of the maize planting area at a 10-m scale in Shandong Province and the comparisons between the estimated maize area versus official statistics in 114 counties for the year 2019 (a, b) and 2020 (c, d).

respectively. This indicates a strong alignment between the estimated maize planting areas and the statistical data in this study.

3. Method

3.1. Data assimilation framework

3.1.1. Soil-Water-Atmosphere-Plant (SWAP) model

SWAP model is a process-based agrohydrological model that could simulate the interactions between soil moisture movement and crop growth (van Dam et al., 1997). The soil moisture movement in the SWAP model is described as one-dimensional vertical, and could be numerically solved by Richards' equation (Eqs. S4) and van Genuchten-Mualem (VG) model (Eq. S5). Five key parameters affect the unsaturated flow simulation in SWAP model, including saturated hydraulic conductivity (K_s), saturated and residual soil moisture (θ_s and θ_r), and the empirical shape factors of VG model (α and n).

The crop growth process involves a number of subprocesses, which are controlled by many different crop parameters (Table S2). For example, the effective temperature sum during the vegetative stage (TSUMEA) and reproductive stage (TSUMAM) have significant effects on the determination of the development stage (DVS), which in turn affects the value of various other parameters, such as specific leaf area (SLA), maximum leaf CO₂ and assimilation rate (AMAX). Thus, it is necessary to identify the most yield-sensitive parameters (Eq. S7) for model calibration.

During the crop growth process, LAI is a crucial crop growth state

variable for yield estimation. LAI values are determined by the combined effects of leaf growth and senescence rates, which are the functions of leaf physiological age, SLA, and leaf dry matter. Among these, the TDWI (i.e., the initial total crop dry weight) and SPAN (i.e., the maximum time in days a leaf can live at 35 °C) parameters directly impact LAI simulations and exhibit significant spatial uncertainties. TDWI indicates the total dry weight of the biomass when the model starts. However, since the actual planting dates are different for various fields at the regional scale, the growth of maize before June 18th has not been considered, leading to significant spatial variability of TDWI parameters. The SPAN parameter controls the rate at which green leaves turn brown, thus determining the crop senescence rate. Although SPAN is characteristic of maize variety, the senescence rate is also influenced by drought, fertilization, and pest/disease infestation. Consequently, the SPAN parameter is affected by the variability in crop management practices at a regional scale. A number of studies have emphasized the importance of these two parameters in crop growth simulations (de Wit et al., 2012; Guo et al., 2024; Ma et al., 2013; Ren et al., 2024).

3.1.2. Iterative ensemble smoother (IES)

IES-SWAP data assimilation framework was developed by Yu et al. (2022, 2020), which aimed to improve the crop growth simulations by fusing various types of measurements into the model. The IES algorithm is a variant of the ensemble Kalman filter (EnKF), which updates the parameters iteratively by assimilating all available measurements. Thus, IES could ensure the temporal consistency between simulations and observations, and has a stronger ability to extract non-linear information

from multi-source data than EnKF. In this paper, the observations of LAI and SSM are assimilated into the model by IES-SWAP framework to estimate both model parameters and yield. A brief introduction of DA settings is presented in Section 3.3.2, and the details about IES-SWAP algorithms are shown in Method S1 in the [supplementary materials](#).

3.2. Machine learning algorithms

Four representative machine learning algorithms were implemented as the emulators of the DA: Feature Tokenizer Transformer (FTT), Artificial Neural Network (ANN), eXtreme Gradient Boosting (XGBoost), and Random Forest (RF). These algorithms were selected because they represent different families of machine learning approaches (deep learning, neural networks, ensemble boosting, and bagging, respectively), thereby providing a comprehensive comparison of predictive capability and interpretability.

FTT is a recently developed deep learning architecture designed for tabular data, where input features are embedded and processed through Transformer encoders to capture complex interactions via self-attention ([Gorishniy et al., 2021](#)). ANN is a feed-forward structure with interconnected layers, capable of approximating nonlinear relationships through backpropagation ([Hornik et al., 1989](#)). XGBoost is a gradient boosting-based ensemble method, favored for its efficiency, scalability, and strong generalization ([Chen and Guestrin, 2016](#)). RF is a bagging ensemble of decision trees that improves stability through bootstrap sampling and random feature selection, while also providing variable

importance measures ([Breiman, 2001](#)).

All four algorithms were achieved using a Python program (version 3.8), and the hyperparameters of each algorithm were optimized using a grid-search method ([Moghimi et al., 2020](#)).

3.3. Yield simulation process based on combined DA-ML method

3.3.1. Scale unification and maize purity extraction

In this section, all regional-scale data are resampled and unified to a 1-km spatial resolution, resulting in a total of 189,401 grid cells (pixels) covering the entire Shandong Province. This standardization ensures consistency across the dataset and facilitates spatial analysis at a finer scale. A 1-km maize purity map was created by overlaying a 1-km grid onto the 10-m maize cultivation map ([Fig. 2](#)) to determine the proportion of summer maize pixels within each grid cell. Additionally, for the regional-scale data with 500-m spatial resolution, such as MODIS LAI and VIs, the pixel aggregation approach was employed to upscale the data to a 1-km grid. For meteorological data and SMAP soil moisture data with an approximate spatial resolution of 10 km, the data were downsampled to a 1-km scale using the nearest neighbor resampling approach.

3.3.2. Yield simulation scheme

The regional yield estimation scheme based on machine learning (ML) requires accurate crop yield data on the regional grids, and should be bounded by physical laws. This study proposed a combined data

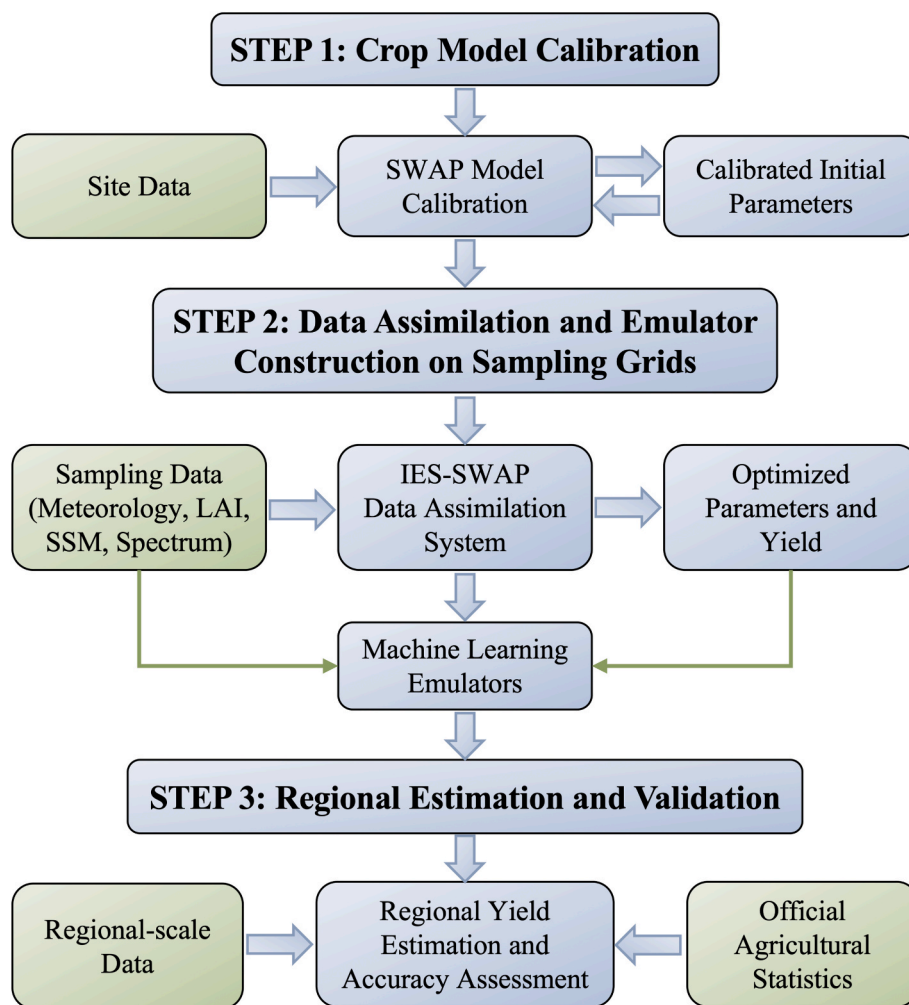


Fig. 3. Flowchart of regional yield estimation process based on combined DA-ML method. The green boxes represent the data type, and the blue boxes represent the simulation process.

assimilation and machine learning (DA-ML) method to address this issue. Firstly, the IES-SWAP model is calibrated using site data to determine the initial model parameters. Then, the model parameters are updated and yield is estimated across numerous (i.e., 100 in this study) randomly-selected pixels (Fig. 1) by merging regional-scale LAI and SSM observations. Afterwards, four ML models are trained to capture the non-linear relationship between regional-scale observations and either crop yield or estimated parameters (from DA method) on these grids. These trained ML models are utilized to predict maize yield and parameters beyond these 100 grids. Finally, the crop yield estimates are verified with official statistical yield data. The flow chart of yield simulation processes is shown in Fig. 3, which consists of three steps:

(1) Calibration of SWAP model and selection of sensitive parameters

A global sensitivity analysis (i.e., Morris method (Morris, 1991)) was conducted to identify the most sensitive parameters of the SWAP model, with the objective of determining the key parameters to be calibrated. Then, the site data in seven experimental plots (i.e., Plot 1 to Plot 7 in Fig. S1) were utilized to calibrate and validate the model. The localized crop parameters were implemented as the initial parameters of the IES-SWAP data assimilation framework. Simultaneously, the soil part of the model was set as a homogeneous soil with a profile of 1 m to simplify the model. The initial soil hydraulic parameter was set as loam, with the value determined based on Carsel and Parrish (1988). In addition, the lower boundary condition was set as a free drainage boundary. Based on the parameter sensitivity analysis results in Section 4.1 and past literature (de Wit et al., 2012; Guo et al., 2024; Ma et al., 2013; Ren et al., 2024; Shi et al., 2015), two crop parameters (SPAN and TDWI) and three soil parameters (K_s , α , and n), which were highly sensitive to LAI and SSM respectively, were selected as unknown parameters in the regional scale. Subsequently, these five key parameters were inversely by assimilating the MODIS LAI and SMAP SSM observations. Detailed descriptions and initial ranges of crop growth parameters used for global sensitivity analysis in the SWAP model are shown in Table S2.

(2) Maize parameter and yield estimations using IES-SWAP on sampling grids

To minimize the influence of other land cover types, the grids with a maize purity exceeding 50 % (Section 3.3.1) were randomly selected for the assimilation process (Huang et al., 2015b). The number of sampling grids is represented by N1 which is set as 100 in this study, and the spatial distribution of sampling grids is presented in Fig. 1. On each grid, the ERA5 meteorology data are utilized to drive the SWAP model, and the MODIS LAI and SMAP SSM data on that grid (Table 1) are assimilated into the model to predict maize parameters and yield. To balance the accuracy and efficiency of IES-SWAP, an ensemble size of 25 with a max iteration number of 10 is determined in this study according to the results of Yu et al. (2020). Additionally, the observation errors of MODIS LAI and SMAP SSM were set to $0.4 \text{ m}^2/\text{m}^2$ and $0.05 \text{ cm}^3/\text{cm}^3$ based on the accuracy of the sensors. Finally, crop yield as well as parameter estimates, that are physically consistent with temporal changes of LAI and SSM, could be obtained on N1 sampling grids.

(3) Construction of ML emulators as a DA substitute for regional yield and parameter estimation

Based on the data assimilation results on 100 sampling grids, four ML regression models are constructed to predict crop yield, SPAN parameter, TDWI parameter respectively. The training output data are the estimated values from the IES-SWAP model in step (2). The input data include all available MODIS LAI observations at different dates, and mean soil moisture (Mean SSM), NDVI (Mean NDVI), EVI (Mean EVI), NDWI (Mean NDWI) observations over the whole growth stage. Importantly, the influence of meteorological conditions during different

maize growth stages was also considered. For this purpose, the entire maize growing season in 2020 was divided into three phenological stages: emergence stage (ES, June 15 to July 15), jointing stage (JS, July 16 to August 7), and reproductive stage (RS, August 8 to September 30). For each stage, average values of net radiation (Mean Radiation), maximum temperature (Mean MaxTem), minimum temperature (Mean MinTem), mean temperature (Mean Tem), dew point temperature (Mean DewTem), wind speed (Mean Wind), and rainfall (Mean Rain) were included as input variables.

70 % of the sampling grids were selected as the training set for the ML models, and the remaining 30 % were used as the testing set. The training set was then subjected to five-fold cross-validation to identify the optimal model, which was evaluated using the testing set for yield estimation. Finally, the trained ML models were applied to each grid to obtain overall crop yield and parameter estimates for the whole region. As MODIS LAI values within each grid may encompass various crops, we made the assumption that the actual maize yield for each grid was the product of the total crop yield and the maize purity coefficient (as computed in Eq. S10). Finally, the estimated regional yield results were validated with the official statistical yield data.

To provide a clear overview of the datasets utilized in each step, a summary of data usage is presented in Table 2.

3.4. Error statistics

In the present study, three quantitative indices R^2 (coefficient of determination), $RMSE$ (root mean square error), and $NRMSE$ (normalized root mean square error) are utilized to evaluate model performances, which are given as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (2)$$

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}} \quad (3)$$

where y_i is the observation; $\hat{y}_i$ represents the estimated value; $\bar{y}$ indicates the average value of the observations; n indicates the number of the observations or estimations; y_{max} and y_{min} indicate the maximum and minimum value of observations.

4. Results

4.1. Crop model calibration

The global sensitivity coefficients of soil and crop parameters for LAI and SSM based on the Morris method (Morris, 1991) are shown in Fig. 4. The analysis indicates that crop parameters of the SWAP model have greater impacts on LAI, while soil parameters play a more significant role in determining SSM. Among the crop parameters (with detailed definitions provided in Table S2), TSUMEA, TDWI, SPAN, SLA, KDIF, EFF, TMPF and conversion efficiency (CVL, CVR and CVS) are important

Table 2
Summary of data usage at different steps.

Step	Dataset description
Step 1	In-situ observations collected from seven plots at Yucheng Station, with data from 2020 used for calibration and data from 2019 used for validation.
Step 2	Regional-scale meteorological, LAI and SSM data in N1 sampling grids
Step 3	Data assimilation results in N1 sampling grids and regional-scale data over the whole Shandong Province

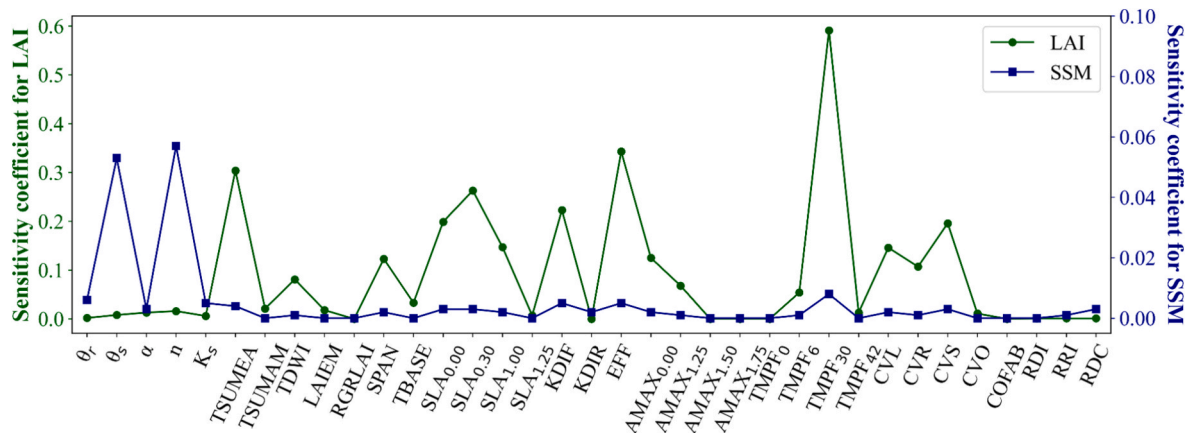


Fig. 4. The sensitivity index of the SWAP model parameters to LAI and SSM (at a depth of 5 cm) based on the Morris method. The definition and initial ranges of crop growth parameters used for global sensitivity analysis are presented in Table S2.

in LAI simulations. While in the soil parameters, the θ_s and n are the most sensitive parameters to the SSM. Other parameters, such as the θ_r , α , and K_s , also exhibit a relatively high sensitivity. To simulate maize growth

processes accurately, these parameters have been calibrated using the site data from seven plots (i.e., Plot 1 to Plot 7 in Fig. S1) or by referring to previous studies (Jiang et al., 2016; Zhou et al., 2013).

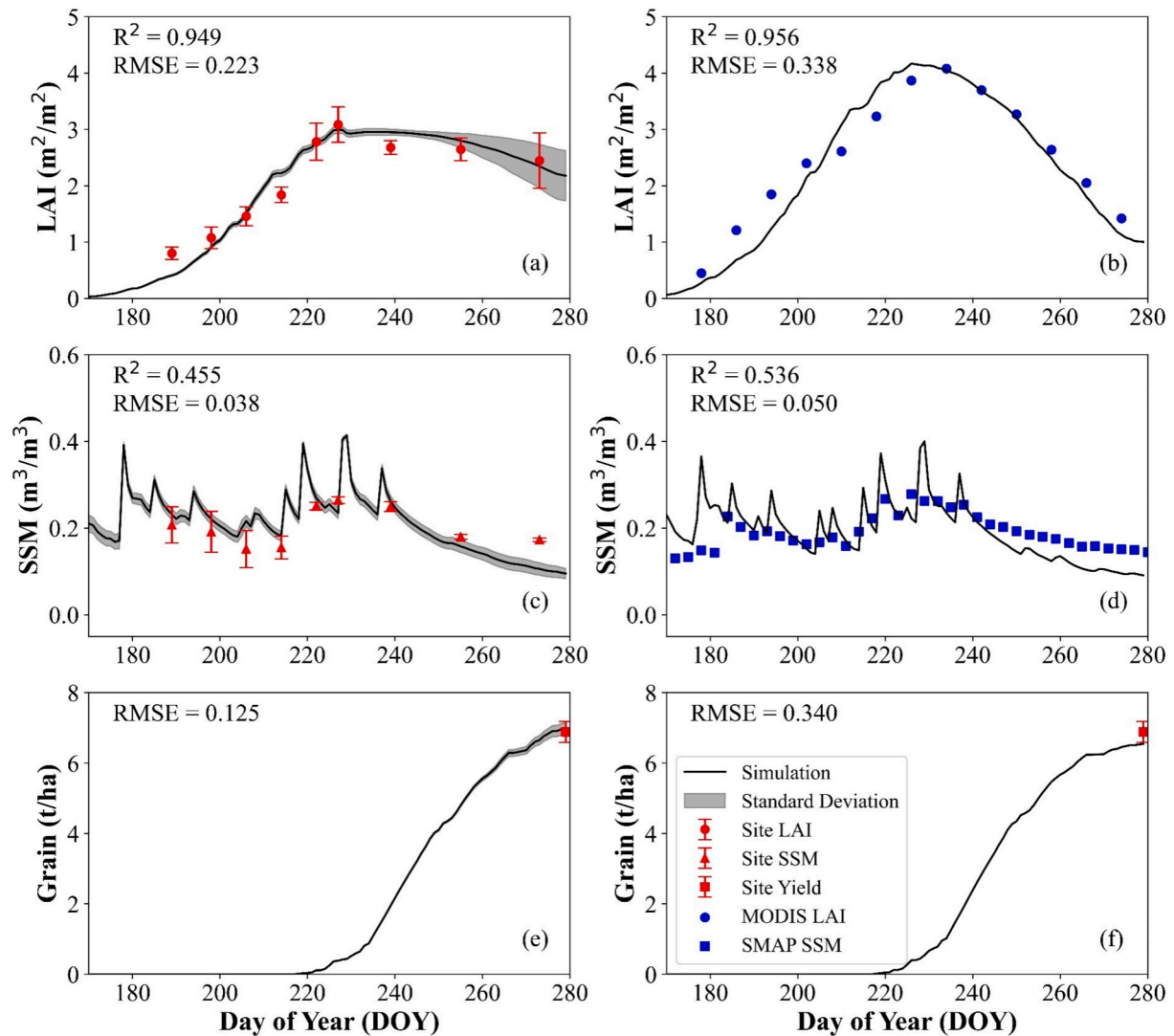


Fig. 5. Comparison of data assimilation results by assimilating LAI and SSM into the SWAP model in 2020. Fig. 5(a), (c), and (e) indicate the mean and standard deviation of simulations by assimilating in-situ LAI and SSM in 7 plots at Yucheng Station (Fig. S3). Fig. 5(b), (d), and (f) present data assimilation results by assimilating satellite MODIS LAI and SMAP SSM data on the corresponding grid.

According to global sensitivity analysis and previous studies (de Wit et al., 2012; Guo et al., 2024; Ma et al., 2013; Ren et al., 2024; Shi et al., 2015), two crop parameters (i.e., TDWI and SPAN) and three soil parameters (i.e., K_s , α and n) are selected to be updated during the assimilation procedure. The data assimilation results by incorporating in-situ LAI and SSM measurements in 2020, and the validation results in 2019 for Plot 1 to Plot 7, are shown in Fig. S3 and S4, respectively. Generally speaking, with the same initial parameters, the yield estimation with the data assimilation method performed well across all seven plots, with RMSE values all below 0.72 t/ha (Fig. S3). In contrast, the validation results for maize growth simulations in 2019 showed some deterioration (Fig. S4), which might be due to differences in meteorological conditions and agricultural management between the two years. Despite higher RMSE values in 2019 compared to 2020 for each plot, the estimations of LAI and yield still matched the observations, indicating the robustness of the SWAP model. In summary, the calibrated SWAP model effectively simulates the maize growth process across different plots and years, confirming its reliability and robustness.

To summarize the data assimilation results in Fig. S3, the mean and standard deviation of the simulations and observations across seven plots are plotted in Fig. 5 (Fig. 5a, c, and e). Additionally, the data assimilation results using MODIS LAI and SMAP SSM data on the corresponding grid are also compared in Fig. 5 (Fig. 5b, d, and f) to evaluate the uncertainty of remote sensing data in model simulations. In general, with the incorporation of LAI and SSM information, the simulated LAI and SSM from the IES-SWAP data assimilation system agree well with observations at site scale (Fig. 5a and c) and regional scale (Fig. 5b and d), indicating that the IES-SWAP model could well characterize the dynamics of crop growth. Compared to site-measured LAI, MODIS LAI exhibits a higher peak, with a steeper increase during the vegetative stage and a sharper decrease during the reproductive stage (Fig. 5b). This discrepancy can be attributed to spatial scale mismatch between field measurements and remote sensing estimates (Fang et al., 2019). The higher peak point might be attributed to the effects of other plant types (e.g., trees) surrounding the Yucheng Station. Additionally, the divergence observed in the late growth season between the site-measured LAI and MODIS LAI is likely due to the different harvesting dates, which are fixed at the site but vary within the 1-km cell. Nonetheless, the calibrated SMAP SSM shows good agreement with in-situ SSM in terms of temporal change and similar mean values (Fig. 5c and 5d). Therefore, the simulated SSM by assimilating calibrated SMAP SSM is capable of providing valuable information on water content.

With respect to yield simulation, since the crop parameters are calibrated based on site data, the yield estimates by assimilating site-specific data could achieve the best results with a mean root mean square error (RMSE) value of only 0.125 t/ha (Fig. 5e). In contrast, for the IES-SWAP model driven by ERA5-Land meteorology data and assimilating MODIS LAI and SMAP SSM data, the RMSE value between

estimated and field-measured yield is 0.340 t/ha, which is also close to the actual measurements (Fig. 5f). This result indicates that although the yield simulation results would deteriorate by using regional-scale data, the physical model can effectively impose constraints on RS observations, leading to more reasonable yield outputs.

4.2. Data assimilation and emulator construction on sampling grids

4.2.1. Data assimilation results

Following the procedure described in Section 3.3.2, 100 grids with a maize purity exceeding 50 % are randomly selected from the maize cultivation map (Fig. 1). Then, data assimilation simulation is conducted on these grids by incorporating MODIS LAI and SMAP SSM data. In Fig. 6 (a) and (b), we present regression results for estimated LAI and SSM in comparison to corresponding regional-scale data. Overall, the information of MODIS LAI and SMAP SSM are successfully incorporated into the IES-SWAP model. The R^2 value for estimated LAI against MODIS LAI is 0.80 with a slope of 0.93, indicating strong consistency between those variables. However, the R^2 value between the estimated SSM and SMAP SSM is 0.46, which is lower than that for LAI simulations. This could be attributed to the coarse spatial resolution of SMAP SSM (10 km), which introduces insufficient soil moisture information into the model, leading to inaccurate estimations of surface soil moisture. Additionally in Fig. 6 (c), we plot the probability density function for the estimated yield generated by the IES-SWAP model. As shown in the figure, the yield estimations exhibit Gaussian characteristics with mean values around 5.6 t/ha, which has a similar pattern in reality.

4.2.2. Comparison of yield predictions using multiple ML emulators

Based on the yield estimation results from the IES-SWAP model (Section 4.2.1), four machine learning models (i.e., FTT, ANN, XGBoost and RF) were developed to establish regression relationships between input variables—including MODIS LAI, SMAP SSM, VIs, and meteorological data (see Section 3.3.2 for details)—and DA yield estimates. The yield outputs from these models were then compared against the DA yield estimates for both training and testing datasets, as summarized in Table 3.

Table 3

Summary of model performances in training and testing datasets.

Model	Train R^2 (—)	Train RMSE (t/ha)	Train NRMSE (%)	Test R^2 (—)	Test RMSE (t/ha)	Test NRMSE (%)
FTT	0.967	0.136	4.533	0.916	0.215	7.167
ANN	0.973	0.119	2.967	0.910	0.223	7.433
XGBoost	1.000	0.013	0.433	0.869	0.254	8.467
RF	0.884	0.258	8.600	0.928	0.214	7.133

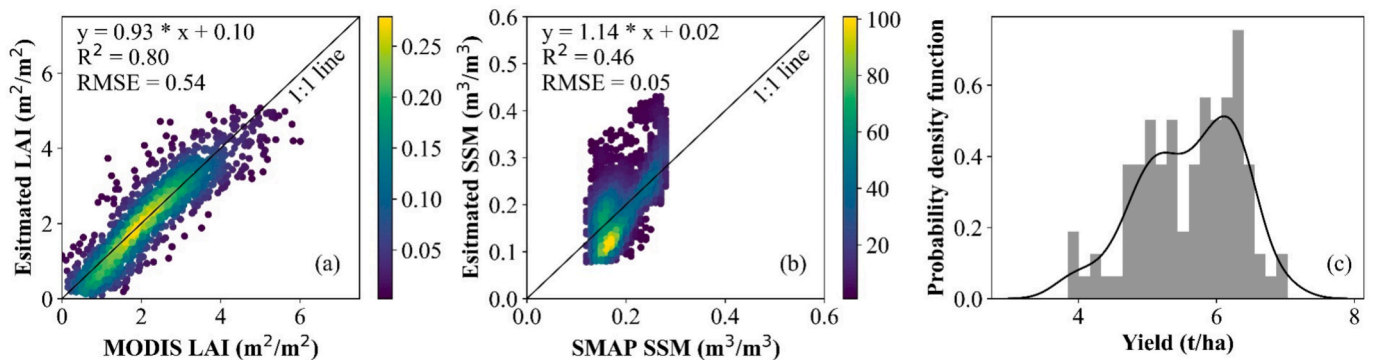


Fig. 6. Data assimilation results by assimilating MODIS LAI and SMAP SSM observations on 100 sampling grids in 2020: (a) simulated LAI versus observations; (b) simulated SSM versus observations; and (c) the probability density function of simulated maize yield. It should be noted that there are a total of 13 MODIS LAI and 37 SMAP SSM data for each grid, and the color bar indicates the density of data points.

As shown in Table 3, the FTT and ANN models demonstrated comparable performances, with testing R^2 values of 0.916 and 0.910, respectively. In contrast, XGBoost achieved nearly perfect fitting on the training dataset ($R^2 = 1.000$, $RMSE = 0.013$) but exhibited a noticeable decline in testing performance ($R^2 = 0.869$, $RMSE = 0.254$), indicating potential overfitting. Although the RF model showed the lowest R^2 value on the training dataset, its comparable accuracy across the training and testing datasets highlights a stronger generalization capability than the other models. This robustness may be partly due to the limited sample size, as more complex models such as FTT, ANN, and XGBoost typically require larger training datasets to fully leverage their learning capacity.

4.3. Regional estimation and validation

4.3.1. Comparison of computation time

All numerical experiments in this study were performed on a Dell Precision 3561 workstation (Intel Core i7-11850H, 8 cores @ 2.5 GHz, 32 GB RAM, no GPU acceleration). In this study, a total of 189,401 1-km grids were analyzed across Shandong Province (Fig. 1). Using the traditional data assimilation (DA) approach, simulating each grid requires approximately 10 min, leading to a total computation time of about 1,315 days—or roughly 3.6 years—if performed sequentially on a single computer without parallel processing. In contrast, the proposed DA-ML framework applies the data assimilation algorithm to only 100 randomly selected grids (Fig. 1), while the remaining grids are processed using a RF surrogate model, which incurs negligible computational cost. Consequently, the total computation time is drastically reduced from approximately 3.6 years to just 16 h, achieving a computational time saving of over 99.8 %. A comparison of the number of grids used for data assimilation and the corresponding computation time between the DA and DA-ML methods is illustrated in Fig. 7.

4.3.2. Pixel-level yield mapping

The estimations of maize yield across the whole Shandong Province for the calibration year (2020) and validation year (2019) are presented in Fig. 8. The pixel-level yield map (Fig. 8a and 8b) highlights that the primary planting areas of summer maize are situated in the northern and western regions of Shandong. Notably, the northwest area, encompassing Dezhou, Binzhou, and Liaocheng municipal districts, serves as a high-yield area for summer maize. While Heze municipal district has the largest planting area of maize, its average yield is slightly lower than that of Dezhou. These distribution patterns closely resemble the corresponding real-world statistical data (Table S1).

To further validate the accuracy of regional yield estimates, Fig. 8c and 8d compare the estimated total yield (Y_{total} , as calculated in Eq. S10) for 114 counties versus the official statistics, based on predictions from the FTT, ANN, XGBoost and RF models respectively. Owing to the spatial uncertainties in agricultural management practices and official statistics, the performance of the ML models deteriorates relative to that observed in the training and testing datasets (Section 4.2.2), which is reasonable given the regional-scale heterogeneity. However, RF model consistently outperforms the other machine learning models in regional yield estimation for both calibration and validation years. Specifically, the RF model achieved $R^2 = 0.62$ and $NRMSE = 18\%$ in the calibration year, whereas FTT, ANN, and XGBoost produced lower accuracies ($R^2 \approx 0.50$, $NRMSE \approx 22.2\%$). Although the prediction accuracy declined in the validation year, the RF model continue to outperform the other models, further highlighting its robustness and practical utility in large-scale yield estimation.

4.3.3. Pixel-level parameter mapping

Similar to the regional yield simulation process, the regression relationship between input variables and crop growth parameters (i.e., SPAN and TDWI) is also constructed on the sampling grids. Then, this relationship is utilized to estimate these parameters across Shandong Province, as presented in Fig. 9. In general, the spatial distribution of SPAN and TDWI parameters significantly affect the spatial pattern of summer maize yield (Fig. 8). SPAN may have more important effects on yield estimation than TDWI parameter. The estimated SPAN parameters show a trend of higher values in the northern regions and lower values in the southern regions (Fig. 9a), demonstrating a strong correlation with the summer maize yield distribution in Shandong. In contrast, TDWI displays a trend of higher values in the southern regions and lower values in the northern regions (Fig. 9b). This difference could be attributed to the fact that TDWI is related to the initial growth stage, while SPAN is related to the leaf senescence rates. In this study, maize grown in southern regions—benefiting from more abundant heat (Fig. 9c) and rainfall (Fig. 9d)—typically exhibits faster initial growth followed by more rapid senescence. This growth pattern results in larger estimated TDWI and lower estimated SPAN values in the south compared to the north, which may explain the spatial distribution in Fig. 9(a) and 9(b).

4.4. Relative importance of inputted features

To further evaluate the data worth of input variables, Fig. 10 presents

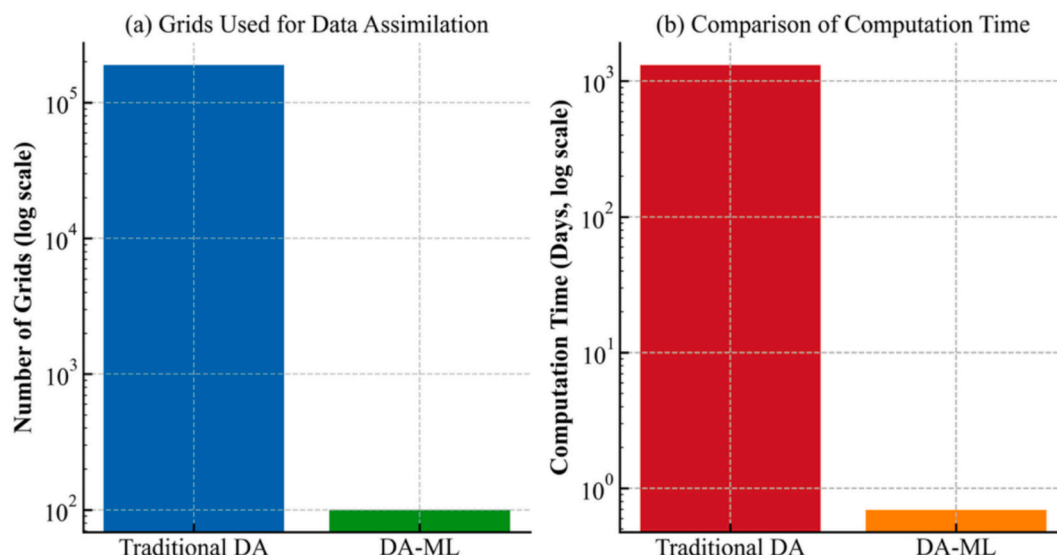


Fig. 7. Comparisons of (a) grid numbers used for data assimilation and (b) computation time between traditional DA method and DA-ML method.

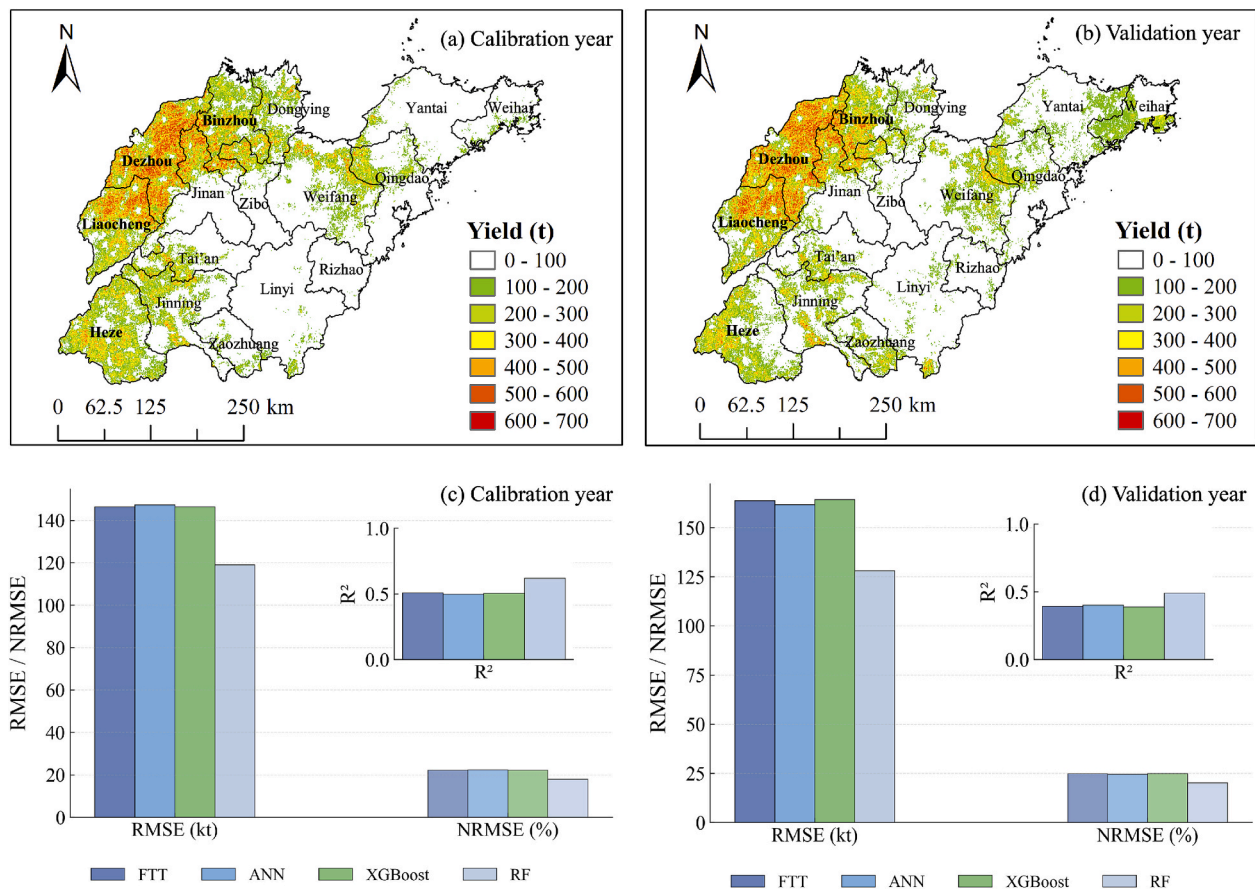


Fig. 8. Simulated (a) summer maize yield with a 1-km scale using RF model, and (b) accuracy of county-level total maize yield (y_{total} , t) across 114 counties in Shandong Province for the calibration year (2020) and validation year (2019), based on four machine learning models.

the relative importance of input variables generated by the RF model. Among the variables in Fig. 10, mean radiation during the jointing stage (Mean Radiation JS) emerged as the most significant predictor, followed closely by mean soil moisture (Mean SM) and mean rainfall during the jointing stage (Mean Rain JS). These top variables underscore the crucial role of light availability, water supply, and precipitation during the key development period in determining maize growth states.

Additional highly ranked variables included mean wind speed (Mean Wind RS) and mean dew point temperature (Mean DewTem RS) during the reproductive stage, suggesting that microclimatic stressors such as wind and atmospheric dryness also substantially impact yield. Temperature variables in the reproductive stage (such as Mean MinTem RS and Mean Tem RS) also showed high importance, indicating sensitivity of the yield response to thermal conditions during late growth phases.

Conversely, variables with lower relative importance included mean rainfall during the emergence stage (Mean Rain ES), leaf area index (LAI) at early dates (e.g., 06–25, 07–11), and mean wind speed at earlier stages (Mean Wind JS), suggesting that early-season conditions, while still contributory, were less decisive than conditions during later phenological stages.

Overall, the variable importance profile highlights the RF model's capacity to capture complex, stage-specific interactions between environmental factors and maize growth, with later developmental stages (especially the reproductive stage) playing a dominant role in final yield prediction. This insight illustrates that the RF algorithm can learn the simulation characteristics and essential physical principles of the IES-SWAP model, making it a suitable surrogate model for rapid regional yield estimation.

5. Discussion

5.1. The potential application of DA-ML framework

Data assimilation (DA) is a classical and widely-used approach for regional crop yield estimation (Jin et al., 2018). Previously, a number of studies aimed to improve the accuracy of regional yield estimation by developing the state-of-the-art algorithm, from ensemble Kalman filter (EnKF) (Evensen, 1994), to ensemble square root filter (EnSRF) (Whitaker and Hamill, 2002), and subsequently to EnSRF with a variable time window (VW-4DEnSRF) (Wu et al., 2021). And the corresponding R^2 values between ground-measured and estimated yield are reported to increase from 0.302 to 0.323, and then to 0.513 respectively, when applying these algorithms to estimate crop yield at a regional scale (Wu et al., 2021). However, with the increase in algorithm complexity, the computational cost of data assimilation will also increase, making it difficult to apply over large areas.

In contrast, known problems in machine learning (ML) involve the challenge of collecting a large amount of high-quality data (Chlingaryan et al., 2018). Real observations are sparse and irregular, with a complex, indirect, and ambiguous dependence on the geophysical state (Geer, 2021). Traditionally, the outputs of physical models could be used as a substitute for the observations (Paudel et al., 2021). However, the uncertainties of crop growth parameters have not been carefully considered at the regional level. Furthermore, DA utilizes observation operators and Bayesian inverse methods to optimize the geophysical state in the physical model. By fusing the information of remote sensing data (e.g., MODIS LAI and SMAP SSM) into the crop growth models, DA system could provide model outputs (e.g., yield data) and parameter estimation (e.g., SPAN and TDWI) in numerous different scenarios.

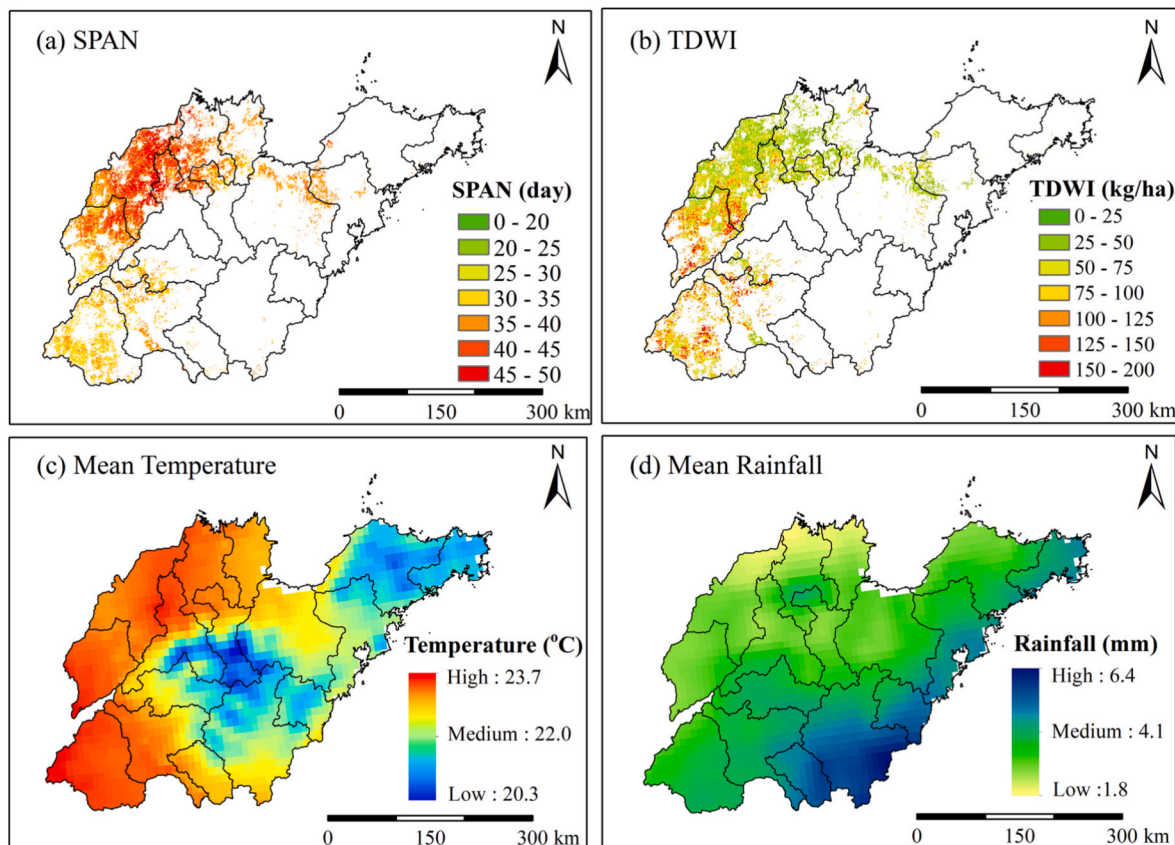


Fig. 9. Estimated 1-km scale (a) SPAN and (b) TDWI parameters using RF model as well as the corresponding mean (c) air temperature and (d) rainfall data during the whole growth period in Shandong in 2020. It should be noted that (a) and (b) only presented the parameter estimations on the grids where “purity” exceeds 50 %.

Therefore, we proposed a DA-ML framework to develop an efficient regional yield or parameter estimation scheme. The total computation time is drastically reduced from about 3.6 years for the traditional DA method to only 16 h for the proposed DA-ML method, achieving a computation time saving of over 99.8 % (Fig. 7). At the same time, the DA-ML model established in this study exhibits robust performances both in calibration and validation years. In 2020, the RF model achieved an R^2 of 0.62 and $RMSE$ of 1.19×10^5 (t) when comparing simulated yields with official statistical data in Shandong Province (Fig. 8c). For 2019, the corresponding R^2 and $RMSE$ were 0.49 and 1.28×10^5 (t) respectively, based on RF model (Fig. 8d). These results are comparable to the maize yield estimations reported by Zhao et al. (2015) and Zhang et al. (2021), demonstrating that the DA-ML method can maintain yield estimation accuracy while significantly reducing computation time. This finding has great potential for precision agriculture applications. With the low-cost estimates of regional crop yield, we can realize real-time agricultural diagnosis, decision-making and management.

Moreover, to further investigate the effects of increased sample grids on regional yield estimation, the RF model was trained using datasets from 500 sample grids, and subsequently applied to Shandong Province to estimate the regional maize yield. The data assimilation results and estimated county-level yields based on 100 and 500 sample grids are compared in Fig. S5 and Fig. S6, respectively. Generally speaking, the RF model trained with 100 sample grids performs similarly to the model trained with 500 sample grids. At the regional scale, the R^2 value between the estimated and statistical total yield across 114 counties are 0.62 and 0.58 for 100 and 500 sample grids, respectively, with the $RMSE$ value remaining the same at 1.19×10^5 (t), suggesting similar accuracy in yield estimation (Fig. S6). Therefore, in this study, increasing the sample size may not enhance the accuracy of regional yield estimation, and 100 sample grids are sufficient to train the RF model effectively.

5.2. The physical principles learned by machine learning method

As indicated by Karniadakis et al. (2021), the observations are able to reflect the fundamental physical principles that govern their generation, and could be utilized as a weak mechanism for incorporating these principles into ML models during its training phase. By using “virtual” yield datasets generated by the DA system to train ML model, our results also agree with this conclusion. Fig. 10 presents the relative importance of input variables generated by the RF model, and shows that the the most influential variables were radiation, soil moisture, and rainfall during the jointing stage, underscore the crucial role of light availability, water supply, and precipitation during the key development period of maize. These findings are consistent with the “physical principles” embedded in the SWAP model. Specifically, LAI (or NDVI) and radiation determine the potential photosynthesis ability of maize, while the soil moisture and rainfall provide insights into crop evapotranspiration and environmental water availability, both of which play a significant role in maize yield formation. This physical principle has been addressed in extensive previous literature (Chen et al., 2013; de Wit et al., 2005; Riha et al., 1996), and is consistent with the assumptions in numerous crop growth models (Hsiao et al., 2009; Jones et al., 2003; van Dam et al., 1997).

In addition, the distribution of SPAN and TDWI parameters can also be estimated with machine learning models, as shown in Fig. 9. The SPAN parameter means the maximum time in days a leaf can live at 35 °C, which is the key parameter indicating the physiologic aging and senescence process (Gilardelli et al., 2018). TDWI denotes the initial total crop dry matter, which is a significant initial variable for crop growth modeling (Huang et al., 2015b). Notably, the estimation of TDWI parameters might be influenced by meteorological data during the early maize growth stage. High temperatures and abundant rainfall after

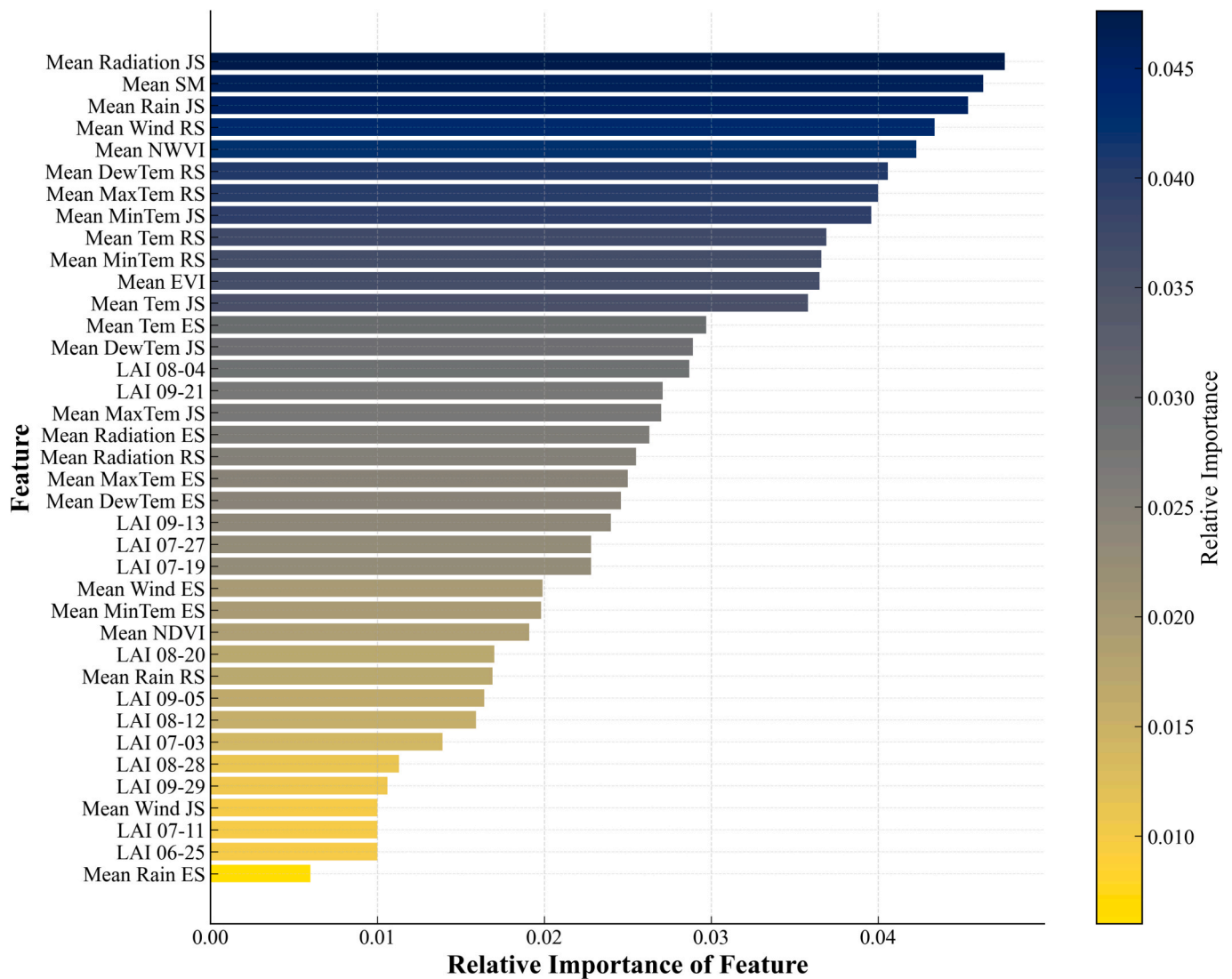


Fig. 10. Relative feature importance of the RF model for maize prediction in 2020.

planting would contribute to the rapid initial growth of maize, resulting in a large TDWI value (Fig. 9). Compared to TDWI, the distribution of the SPAN parameter is more consistent with the distribution of crop yield (Fig. 8). This result is reasonable since a slower leaf senescence rates would contribute to a larger SPAN, thereby enhancing yield estimation in the RF model. Interestingly, according to the global sensitivity analysis results from this study (Fig. 4) and previous literature (Hu et al., 2017), the sensitivity coefficient of the SPAN parameter to LAI or yield is also higher than that of the TDWI parameter in the SWAP model, indicating a comparable physical law between physical model and machine learning. In summary, all of the above results demonstrate that the constraining physical principles could be incorporated into the ML model by involving DA outputs into the ML training phase.

Additionally, the mapping of soil parameters (K_s , α , and n) is not presented in this study, as the constructed ML model only considers the Mean SSM, which could result in potentially inaccurate outcomes. Previous research by Araya and Ghezzehei (2019) suggests that incorporating soil structure information, such as the 10th percentile particle diameter, clay content, bulk density, and organic carbon content, into machine learning models is essential for accurately predicting K_s . Therefore, how to precisely estimate regional soil parameters by using a machine learning model is another interesting but challenging problem, which should be investigated further in the future.

5.3. Uncertainty and potential refinement

It is important to recognize that the current study has some uncertainties and limitations. Firstly, the accuracy of IES-SWAP model outputs significantly affects the reliability of ML model. Although the IES-SWAP model is carefully calibrated based on the site data, many influencing factors have not been adequately explored when employing this model on the regional grids. Regional yield variability is a result of complex interactions among hybrids, farmer management practices (e. g., density, fertilizer and irrigation), soil properties, and weather conditions. In this study, we used ERA5-Land meteorology data to account for spatial uncertainty of weather conditions, but its resolution (i.e., 10 km) is too coarse to capture changes at the farmland level. Additionally, the spatial uncertainty of soil properties was addressed by inverting soil parameters (K_s , α , and n) for each grid, which is a simplification when applied on a large scale. The assumption of vertical homogeneity in the soil layer further adds to the uncertainty in soil moisture simulations. Moreover, collecting data on hybrids and farmer management practices at a regional scale is challenging. While regional irrigation information can be indirectly incorporated into the model by assimilating SSM data (El Sharif et al., 2015), assumptions about uniform planting density and optimal fertilizer application across large areas introduce additional bias in yield estimates. In summary, these factors contribute to uncertainties

in IES-SWAP model predictions, warranting further investigation.

Secondly, the introduction of ML model would inevitably introduce the uncertainties of model structure, resulting in the biased estimation of yield outputs. In this study, FTT, ANN, XGBoost, and RF algorithms have been employed to learn the physical patterns from DA system in this study. Although FTT, ANN and XGBoost are advanced ML methods, the limited size of input data may constrain their performances, allowing the RF model to outperform them in regional yield estimation. Recently, physics-informed machine learning methods, which embed physics into machine learning, are attracting more and more attention. The primary motivation for developing these algorithms is to make ML methods produce physically-consistent and exact predictions even trained by imperfect data (such as noisy or missing values) by using prior knowledge or constraints (Karniadakis et al., 2021). Given the substantial uncertainties in remote sensing data, these methods may be particularly well-suited for applications within the field of remote sensing. Thus, it is essential to assess and compare current model performances against these advanced models to select the optimal ML method in the future. In addition, the types of input data used in this study were also limited. To assess the feasibility and effectiveness of ML model, we relied minimal yet widely accessible set of inputs (e.g., LAI, soil moisture, and meteorology data) similar to those commonly employed in process-based crop models. These inputs are derived from widely available optical satellite products and reanalysis datasets, which are relatively easy to acquire and process. However, several recent studies have demonstrated the benefits of embedding additional data types such as chlorophyll fluorescence, synthetic aperture radar, optimal and thermal remote-sensing data (Kira et al., 2024; Lu et al., 2025; Veloso et al., 2017), which should be investigated further in the future.

Finally, the “mixed pixel” has significant effects on data assimilation results and yield estimation, which is still a challenging problem in the remote sensing field. In this study, we assume that grids with maize “purity” above 50 % would be used in the DA system, following the approach of Huang et al. (2015a). However, the validity of this threshold as an optimal criterion for crop type representation remains uncertain. In particular, the method’s applicability is constrained by the smallholder farming structure prevalent in China (Zhang et al., 2016). Unlike large-scale commercial agriculture, smallholder systems typically feature highly fragmented fields, diverse intercropping patterns, and irregular planting schedules. These characteristics create a complex mosaic landscape where a single 1-km MODIS pixel may contain numerous small fields with different crops, growth stages, and management practices. This heterogeneity significantly reduces the “purity” of crop-specific signals within each pixel, which introduces systematic biases in crop growth monitoring and subsequent yield estimations. To address these limitations, future research could explore the integration of multi-source remote sensing data with higher spatial resolution, the development of advanced mixed-pixel decomposition techniques, or the implementation of region-specific calibration strategies tailored to the unique features of smallholder agriculture.

6. Conclusion

In this work, we presented a DA-ML framework to achieve an efficient and physics-informed regional yield (or parameter) estimation scheme. We initially conducted the global parameter sensitivity analysis of the SWAP model, and calibrated the SWAP model using the site data in Yucheng Station. Then, the “virtual” maize yield (or parameter) observations on a number of randomly-selected maize grids were generated by the IES-SWAP system, which fuses the information of MODIS LAI and SMAP SSM data. Moreover, we employed the four machine learning algorithms to establish the regression relationship between remote sensing data and “virtual” yield (or parameter) observations, and applied these machine learning models to achieve regional yield (or parameter) estimates. The regional yield estimation results were evaluated and validated by two-year agricultural statistical data.

Three main conclusions can be drawn from our study:

- (1) Compared to the traditional data assimilation method, the combined data assimilation and machine learning (DA-ML) framework significantly reduces the computational cost for regional yield and parameter estimation while maintaining yield estimation accuracy.
- (2) Compared to outputs from a physical model, incorporating data assimilation outputs into the machine learning training process ensures consistency between observations (e.g., LAI and SSM) and yield outputs.
- (3) For regional maize yield estimation, the most influential variables were radiation, soil moisture, and rainfall during the jointing stage, followed by wind speed, air humidity, and temperature during the reproductive stage, highlighting the dominant role of environmental conditions during later growth stages in determining final yield.

Further research may consider more mechanistic physical models, and evaluate the potential of accurately generating other essential crop growth variables (e.g., ET, GPP, and land surface flux) across the regional area using the proposed method in this study.

CRediT authorship contribution statement

Danyang Yu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuanyuan Zha:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Yijian Zeng:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Peiyu Lai:** Methodology, Formal analysis, Data curation, Conceptualization. **Wanxue Zhu:** Formal analysis, Data curation. **Jiang Bian:** Formal analysis, Data curation. **Qi Yang:** Methodology, Investigation, Data curation. **Xi Huang:** Formal analysis, Data curation. **Zhongbo (Bob) Su:** Writing – review & editing, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2025.111142>.

Data availability

Data will be made available on request.

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